Xuanchen(Jason) Liu

Full-Stack Developer / Data Engineer

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Work Experience

Solutions Developer

Aug 2023 - Present

Hamilton Health Science (Population Health Research Institute)

- Collaborated with study teams to translate dynamic payment criteria into SQL queries, identifying eligible participants based on complex study protocols and mortality scenarios.
- Developed and executed SSIS packages to synchronize data from SQL Server to Oracle databases, ensuring data consistency and reliability across systems.
- Enhanced in-house clinical data management system to support configurable study workflows and data processing logic.
- Collaborated with cross-functional teams to deliver data-driven solutions in an Agile environment.

Full-Stack Developer

Jan 2021 - Jul 2023

Orispark Inc.

- Developed and maintained backend systems using C# (.NET) and Python (FastAPI), focusing on data processing, API design, and system scalability.
- Designed and implemented RESTful APIs to handle data validation, transformation, and integration across systems.
- Worked with relational databases (MySQL) to design schemas, write complex queries, and ensure data integrity.
- Improved system performance through debugging, query optimization, and efficient backend logic implementation.

Technology Researcher & Developer (Co-op)

Sep 2019 - Aug 2020

World Vision Canada

- Developed internal applications using PHP and MySQL, focusing on data storage, retrieval, and reporting.
- Designed and tested database-driven features to support business operations.
- Collaborated with teams using Azure DevOps in an Agile environment.

Projects

Clinical Payment Data Pipeline (Finance Team)

Oct 2024 - Present

- Reconstructed and standardized the payment reporting workflow for clinical studies in the absence of documentation
- Analyzed and refactored legacy SQL scripts to ensure accurate identification of payable patient visits
- Integrated data from multiple clinical systems (TrialMaster - SQL Server, DataFax - MySQL) into a unified payment workflow
- Ensured quarterly payment data consistency by validating and synchronizing visit-level data into the finance payment system
- Collaborated with finance and study teams to translate complex business rules into reliable data processing logic

Core Skills

Languages: C#, SQL, Python, React, Node.js

Database: SQL Server, MySQL, Firebase

APIs: RESTful API development and integration

Tools & Platforms: Azure DevOps, GitHub, Jenkins

Education

Sheridan College

Sep 2017 - Dec 2020

Advanced Diploma Software Development and Network Engineering

Certificates

ITIL® V4 Foundation

Dec 2023

PeopleCert

Managing Big Data with MySQL

Sep 2020

Duke University | Coursera